

Module 16: HTML & CSS

Course Objectives

At the end of this course, you will be able to:

- Understand the Fundamentals of HTML and CSS
- Implement List, Table ,Images elements
- Utilize selector in CSS
- Understand Box Model in CSS
- Implement Responsive Web Design

Introduction to HTML

HTML Overview

- HTML (Hypertext Markup Language) is markup language that use to defines structure of web page.
- Web pages execute in browsers to display output of the webpage.
- Hypertext refers the links that connect web pages to another page.
- HTML tags are case insensitive; however, it recommends to use lower case letters.
- HTML tags are written inside angular braces(<>) & closed with forward slash(/) character.
- Every tag should be closed otherwise contents(Data) will not appear in structure format.
- The World Wide Web Consortium (W3C) is used to define HTML and CSS standards.

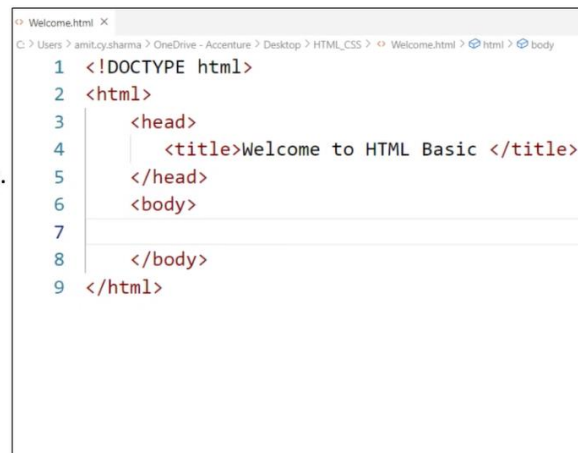
HTML Brief History

HTML developed based on SGML(Standard Generalized Markup Language) by Tim Berners Lee. It was a very basic and some tag such as p, h1, etc. in first version.

Version	Released Year
HTML 1	1993
HTML 2	1995
HTML 3	1997
HTML 4	1999
HTML 5	2014

How to create first HTML page?

- Require a text editor to write the code.
- Saved the file with .html extension in project working directory.
- Every HTML 5 page start with <DOCTYPE html>.
- Every webpage start with <html> and end</html> tag. It is root tag for entire document.
- Web page consist of head and body elements.
- Head element is a container that used to store metadata about the page.
- Body element is used to display the value to user.



```
1 <!DOCTYPE html>
2 <html>
3   <head>
4     <title>Welcome to HTML Basic </title>
5   </head>
6   <body>
7
8   </body>
9 </html>
```

Understanding Elements in HTML

Basic Elements In HTML

Tag	Description
<!DOCTYPE html>	It describe document type. It represent HTML 5 document.
<html>	Every document start and end this tag. It is called root tag in HTML.
<head>	It utilize to define the metadata about the document.
<title>	It defines page title & place inside <head> tag.
<body>	It defines content of document. There can utilize single <body> element in entire document.
<h1> to <h6>	To define level of heading in HTML document. There are six heading elements.
<p>	To define paragraph in HTML document & insert a blank line after the paragraph.
 	It used to break the link in HTML document.
	To insert a horizontal line in HTML document.
<!-- Comments -->	To define the comments in document. Commented Context does not visible in browser.

Some formatting elements

Tag	Description
	To define text content in bold format.
<u>	To define underline below text content in HTML document.
<i>	To define text content in italic format.
<sup>	To define super script text content in HTML document.
<sub>	To define sub script text content in HTML document.

Few Important Elements

Tag	Description
<a>	- It is used to create hyper link to web pages, files, email addresses, locations in html document. Content - Specify URL(Uniform Resource Locator) value in href attribute. - target attribute specifies where you must open linked document. Values are __blank, __self, __top, __parent. It is option attribute & Default value is __self.
<div>	It defines section (block) on HTML document.
	It defines inline container inside the page. It use when you want to apply style sheet on a particular text or element.
	- To add image on html document with heigh and width attribute. - src attribute describes the image path. - alt attribute describe alternative text for image, if image unable to display.

--> We created an app
--> Objects in app, Fields in objects
--> Formula fields, Relationship Fields
--> Validation Rules
--> Data Security --> profiles, Permissions , OWD, sharing rules
--> Flows
--> Data import and export,
--> Reports and Dashboards
Programming section:
--> Apex classes --> basics of OOPs
--> SOQL and SOSL
--> Apex Triggers^I

--> LWC -- > 5 days
--> Asynchronous apex
--> Integration
--> Apex Testing

WE have different sets of HON assessment:

1. Set 1 --> Bike Rental app --> You will be given the whole information of objects, fields that you have to create. Relationship fields, You have to understand.

There are going to be 5 to 6 questions^I that you have to complete which can include to create formula field, create validation rule, create triggers, create flow, create report, create a test class, Creating an LWC(mandatory in all scenarios)

- 2. Set 2 --> Student Management App
- 3. Set 3 --> Employee Management App
- 4. Set 4 --> Bank Loan App

^I
after completing the solutions to these questions, you have to create a solution document and upload it on [mettl](#).

Then you will have the viva. In viva, the questions can be asked from this HON scenario and the solutions you have submitted.

List Elements In HTML

- List serve the purpose of organizing a collection of connected elements. HTML provides three primary list formats:
 - Unordered List
 - Ordered List
 - Description List

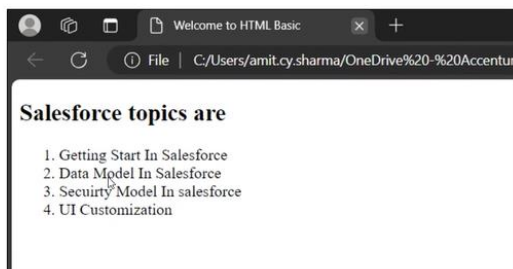
Unordered List()

- It is used for items that do not require to be in any specific order.
- The list items are represented with bullets by default.
- List Item is represented in tag.

```
1 <!DOCTYPE html>
2 <html>
3   <head>
4     <title>Welcome to HTML Basic </title>
5   </head>
6   <body>
7     <h2>Programming Languages are </h2>
8     <ul>
9       <li>Apex</li>
10      <li>Java</li>
11      <li>C++</li>
12      <li>PHP</li>
13    </ul>
14  </body>
15 </html>
```

Ordered List()

- It is utilized when the order of the items is required.
- The list items are represented with numbers by default.
- List Item is represented in tag.



```
1 <!DOCTYPE html>
2 <html>
3   <head>
4     <title>Welcome to HTML Basic </title>
5   </head>
6   <body>
7     <h2>Salesforce topics are </h2>
8     <ol>
9       <li>Getting Start In Salesforce</li>
10      <li>Data Model In Salesforce</li>
11      <li>Secuirty Model In salesforce</li>
12      <li>UI Customization</li>
13    </ol>
14  </body>
15 </html>
```

Demo

Requirement on Basic HTML Elements



Develop basic html page for ABC institute as displayed in right side

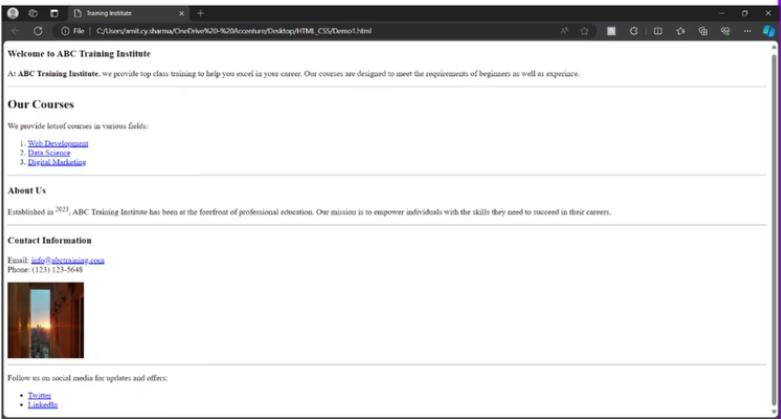


Table Elements In HTML

It utilize to construct a table, which is structured set of data made up of rows and columns.

Tag	Description
<table>	To define table structure on page. Every table start and end with this element. It act as container that contain rows and columns.
<tr>	To define row in table element.
<td>	To define column in table element
<th>	To define heading in table element.
<thead>	To define header context inside the table
<tbody>	To define body

Attributes in Table

Attribute	Description
colspan	This attribute is used to merge multiple columns data into a single column. You can define in <th> or <td> element.
rowspan	This attribute is used to merge multiple rows into a single row. You can define in <th> or <td> element.
border	To define table border on html document.
cellspacing	To define space between cells
cellpadding	To define space between cell and content.
bgcolor	To update table background color, You can define this property in hexadecimal value, rgb value or color Name .
align	To define table alignment on document such as left, right, or center. By default, left align on the document.
height	It is used to define height of table.
width	It is used to define width of table.

Demo

Requirement For HTML Table



Add all employee's information in tabular format on HTML document. To accomplish this requirement, we must implement this screen shot shown in right side.

Employees Details		
Employee Name	Job Title	Department
John Le	Analyst	Sales
Robin Sharma	Sr. Analyst	Finance

<form> Elements In HTML

The <form> element is utilized to create context where users can submit information.

- It contains various type of inputs check as text field, checkboxes radio buttons , text area and mores.
- Syntax:

```
<form action="" method="">
    various form elements
</form>
```

- Action and method are attribute of form element.

<form> Elements In HTML

Tag	Description
<input>	It is used to collect various type of data from user.
<select>	To create drop-down list with <option> element.
<textarea>	To create multi line text area with rows and columns attribute.
<button>	To define clickable button on document.
<label>	To define text label in document.
<datalist>	specifies a list of pre-defined options for an <input> element.
<output>	represent the result of a calculation

<input> Type attributes In HTML

Type Attribute Values	Description
text	text value is used to enter text data in single line.
email	It is used to input email data.
date	It is used to select.
radio	It is used to define radio button on document. Suppose you want to define three radio button such as male , female, and other. All radio button names must be same.
Checkbox	It is used to define a checkbox.
file	It is used to upload file
search	It is used to define “search box”
submit	It is used to define submit button
reset	It is used to define reset button
number	It is used to input numeric data

Form Attributes

Attribute	Description
required	To define mandatory field in HTML form.
readonly	To define read only field
disabled	To disabled the element on HTML form,
size	It can use with <input> and <select> element. It specify number of character visible in <input> element, however it defines number of visible options in drop-down list.
value	To define value for option, input etc.
multiple	It defines that user can select multiple values from drop-down list or upload multiple file using <input> element.
pattern	To define regular expression to match pattern
placeholder	To provide support text inside input box.

Demo

Requirement On HTML Form



Create a HTML form that allows new employees to register their details in company database.

The form should collect below details:

First Name , Last Name

Email

Date of Birth

Gender (With three options male, female, and other)

Contact Number

Department(Drop down List)

Address (Multi Line Area)

Employee Registration

First Name

Last Name

Email

Date of Birth

Gender

☐ Male ☐ Female ☐ Other

Contact Number

Department

Address

Understanding CSS Fundamentals

CSS Overview

- CSS(Cascading style sheet) is defined that how elements should be presented on HTML or XML.
- The W3C specifications ensure that CSS is uniformly standardized across all Web browsers.
- There are three type of CSS.
 - Inline CSS
 - Internal CSS
 - External CSS

Difference among Inline, Internal, and External CSS

Inline CSS	Internal CSS	External CSS
It is applied on element using style attribute. For example: <h1 style="background:red; color:green">	To apply internal CSS, you define all CSS properties inside <style > element that is created inside <head> element. For Example: <style> h1{ background:red; color: green; } </style>	To apply external CSS on page, you define entire CSS properties in external CSS file & file must be saved with .css extension. For Example: <link rel="stylesheet" href="fileName.css">
It overrides both Internal as well as external CSS.	It overrides only external CSS.	It does not override any internal as well as inline CSS

Style Elements in CSS

Text Style

CSS text styling encompasses a range of properties that enable you to customize text format in diverse styles.

Properties	Description
text-align	It is used to align the text to left, right, center, and justify it across the line.
text-decoration	It is used to define various text decoration such as underline, overline, or line-through.
text-transformation	It is used to change text appearance such as uppercase, lowercase, or capitalize

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Border Style

It is used to define an element border.

Properties	Description
border-style	It is used to define various type border such as dotted, dashed, solid, double, none , hidden, or more.
border-color	It is used to define border color in various format.
border-width	It is used to define border width in specific format (px, pt, cm, em, etc)

You can define all property in single line as

```
{border: border-width border-style(required) border-color}
```

You can define one side border property such as

```
{border-left: 6px dotted green}
```

Background Style

It is used to modify the background property of an element.

Properties	Description
background-color	To change background color of an element. For example: {background-color: red;}
background-image	To set image in background of an element. For example: {background-image:url("fileName.extension");}
background-repeat	This property is used to repeat background image in vertical(repeat-x), horizontal(repeat-y), or none. For example,{ background-repeat: repeat-x}
background-position	To set background image start position such as

All properties are defined in single line as: {background: color image repeat position}

Padding Style

- It defines space around element content from its border.
- Padding values can define as px, pt, cm, %, inherit, and mores.

Properties	Description
Padding-left	Define space from left side of content
Padding-right	Define space from right side of content
Padding-top	Define space from top side of content
Padding-bottom	Define space from bottom side of content

- All padding properties are defined in single line as

```
{padding : top right bottom left}
```

Margin Style

- It defines space around element border .
- Margin values can also define as px, pt, cm, %, inherit, and mores.

Properties	Description
margin-left	Define space from left side of element border
margin-right	Define space from right side of element border
margin-top	Define space from top side of element border
margin-bottom	Define space from bottom side of element border

- All margin properties are defined in single line as
{margin : top right bottom left}

Few Essential Properties

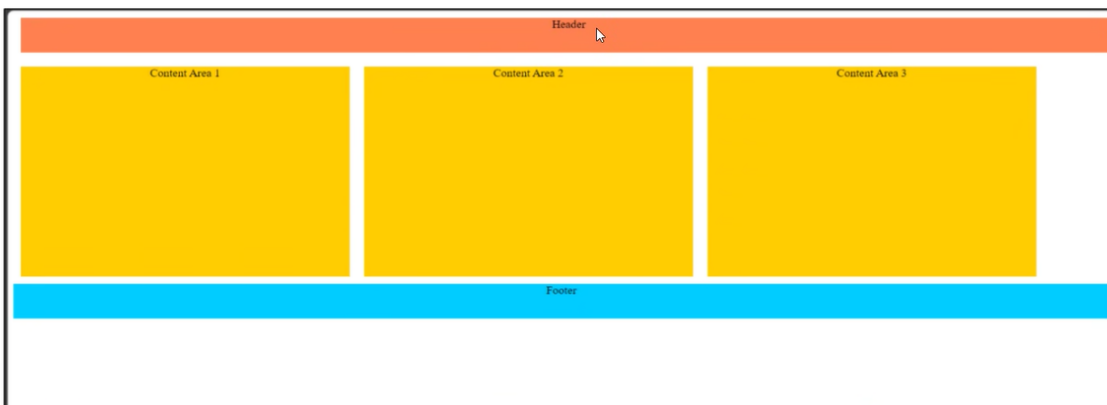
- float: it define to place an element to left , right or none in document.
- clear: it defines if an element must be moved below (clear) floating elements that precede it in the markup.
- position: It defines how element is positioned in a page layout. You can define position as fixed, absolute, relative, static, or sticky.

Demo

Float and Clear Property



- Demonstrate Float and clear property according to this output



What is Selector in CSS?

It is used to target the HTML elements on web pages that require formatting(Styling). Various type of selectors are present in CSS.

- Simple Selectors
- Pseudo Class Selectors
- Pseudo Element Selectors

Simple Selectors

- It is targeted the element by name, id, and class.
- Id and class are attributes of an element.
- Id must be unique attribute on a page whereas class can be utilized in multiple elements as attribute.
- To target an element that has a particular id, you should use a hash (#) symbol, followed by the element's id whereas class should use a dot(.) symbol, followed by the element's class.

Demo

Simple Selector



- Develop sample html page to utilize internal CSS as well as implement Simple selector to style page.

```
Sample.html
1 <!DOCTYPE html>
2 <html>
3   <head>
4     <title>Internal CSS Example</title>
5     <style>
6       h1{
7         color: aqua;
8       }
9       #first{
10        color: blue;
11        font-size: 16px;
12        font-style: italic;
13      }
14      .sample
15      {
16        color: yellow;
17        background-color: green;
18      }
19    </style>
20  </head>
21  <body>
22    <h1>Welcome to HTML & CSS Learning </h1>
23    <p id="first">Learning Simple Selector</p>
24    <p class="sample">First Paragraph</p>
25    <p class="sample">Second Paragraph</p>
26  </body>
27 </html>
```

Pseudo class Selectors

- It are utilized to specify the unique states of an element which simple selectors are unable to target.
- Few common pseudo-classes and their uses:
 - :hover: Format an element when the user's pointer hovers over it.
 - :visited: Format links that the user has visited.
 - :active: format the active link or element that is being clicked.
 - :focus: Format an element like an input field when it has focus.

Syntax:

Selector:pseudo-class{

Property:value;

}

Demo

Pseudo Class Selectors



- Build a sample html page to demonstrate external CSS as well as used pseudo class selectors.

```
1 <!DOCTYPE html>
2 <html>
3   <head>
4     <title>External CSS Demo</title>
5     <link rel="stylesheet" href="sample.css"/>
6   </head>
7   <body>
8     <p>Thank you for selecting this article.
9     If you require more information about this article.
10    <a href="https://www.accenture.com/in-en">Click Here</a>
11  </p>
12 </body>
13 </html>
```

```
1 a:link {
2   color: #FF0000;
3 }
4
5 a:visited {
6   color: #00FF00;
7 }
8
9
10 a:hover {
11   color: #FF00FF;
12 }
13
14
15 a:active {
16   color: #0000FF;
17 }
```

Pseudo Element Selectors

- It are utilized to formatted specific parts of an elements. For instance, you would like to format first letter or line of an element.
- Pseudo-elements and their uses:
 - ::first-letter: Formats the first letter of targeted element.
 - ::first-line: Formats the first line of text of targeted element.
 - ::before: Insert the contents before an elements.
 - ::after: Insert the contents after an elements.
- Syntax:
Selector::pseudo-element{
Property:value;
}

Demo

Pseudo Element Selectors

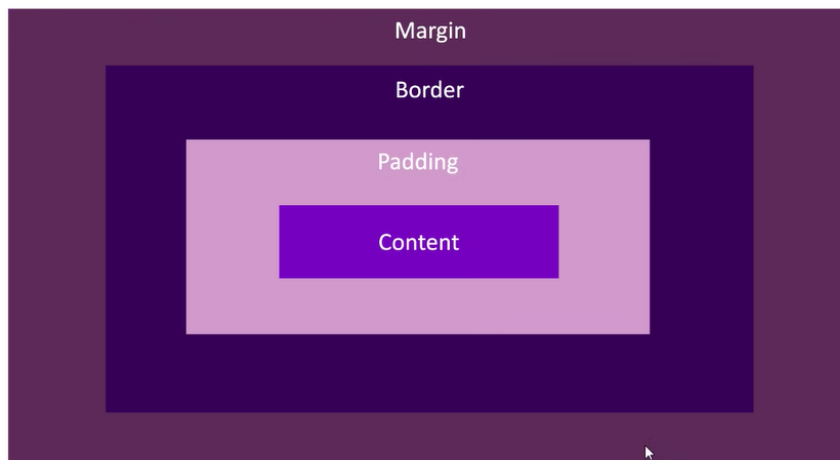


Develop the code to demonstrate pseudo element selectors in CSS.

```
1 <!DOCTYPE html>
2 <html>
3   <head>
4     <title>Pseudo Element Selector</title>
5     <style>
6       p::first-letter {
7         color: red;
8         font-size: 48px;
9         font-family: 'Times New Roman', Times, serif;
10      }
11    </style>
12  </head>
13  <body>
14    <p>Thank you for selecting this article.
15      If you require more information about this article.
16      <a href="https://www.accenture.com/in-en">Click Here</a>
17    </p>
18  </body>
19 </html>
```

Understanding the Box model in CSS

What is Box Model in CSS?



What is media Query?

- It is used to create responsive web page.
- Serve the purpose of implementing varied styles across diverse media types and devices.
- Enables to check various aspects, including the viewport's width and height, the device's dimensions, orientation, and screen resolution

What is media Query?

- Syntax:

```
@media not|only mediatype and (media feature) {  
    /*Define CSS Rules */  
}
```

- Every media query start with @ media.
- Mediatype is option and can print, screen ,all, and mores. If you do not define it by default all.
- "Media Feature" could be height , width and orientation .
- not keyword is used to invert the meaning of entire media query.
- only keyword is used to restrict older browsers that do not support media query. There is no effort in modern browser.
- and keyword is used to merge a media type with one or more media features.

Demo

Media Query



- Previously we created the box layout. Now we have to ensure that content area adjusts dynamically with changes in Screen Size.
- Screen size less than 600px than output of previous code.

```
@media screen and (max-width: 600px) {  
  .topnav {  
    flex-direction: column;  
  }  
  .box {  
    width: 100%;  
  }  
}
```



To-do

Media Query



Create a layout that changes from a four-column layout into a single-column layout on smaller screens.

